BAIO LEARNING SYSTEM

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HIGH-PRIORITY BOARD EXAM TOPICS BLUEPRINT

Chapter: Molecular Basis of Inheritance • Ranked by Recurring Weightage, Concepts & Diagrams

CATEGORY 1: DEFINITE 5-MARK LONG ANSWER ESSENTIALS

Highest Probability • Direct Theory + Diagram

1. Watson-Crick Double Helix Model of DNA

TOP PRIORITY • 5M

- **Salient Structural Features:** Antiparallel polynucleotide strands (5'→3' and 3'→5'), sugar-phosphate backbone, inward base orientation.
- **Complementary Base Pairing:** Strict purine-pyrimidine pairing (A=T with 2 H-bonds, G≡C with 3 H-bonds) ensuring uniform 2.0 nm (20 Å) diameter.
- **Pitch & Stacking:** 3.4 nm (34 Å) pitch, 10 bp per complete helical turn, 0.34 nm (3.4 Å) rise per bp, and stability from hydrophobic base-stacking planes.

Mandatory Board Diagram: Antiparallel strands with polarity, sugar-phosphate chain, base pairs with H-bonds, pitch (34 Å), diameter (20 Å).

2. Regulation of Gene Expression — The Lac Operon

TOP PRIORITY • 5M

- **Operon Composition:** Regulatory gene (*i* gene), promoter (*p*), operator (*o*), and structural genes (*z*, *y*, *a*).
- **Enzyme Mapping:** *z* codes for β-galactosidase, *y* codes for β-galactoside permease, *a* codes for β-galactoside transacetylase.
- **Mechanism of Switching:** Repressor-operator binding in absence of inducer (OFF state) vs. Allolactose/lactose inducer binding to repressor allowing transcription (ON state).

Mandatory Board Diagram: Schematic representation of the Lac Operon cassette under repressed (OFF) and induced (ON) conditions.

3. Hershey & Chase Experiment (Unequivocal Proof)

TOP PRIORITY • 5M

- **Isotope Labeling:** Bacteriophage T2 labeled with ^{35}S (protein coat) vs. ^{32}P (DNA core).
- **Sequential Steps:** 1. Infection of *E. coli* → 2. Blending (stripping viral coats) → 3. Centrifugation (pellet vs supernatant fractionation).
- **Inference:** Radioactivity found in pellet for ^{32}P (proves DNA entered bacteria) vs supernatant for ^{35}S.

Mandatory Board Diagram: Flowchart showing phages infecting *E. coli*, blending stage, and centrifugation result tubes.

4. DNA Fingerprinting Principle & Workflow

HIGH PRIORITY • 5M / 3M

- **Principle:** High-degree polymorphism in non-coding satellite DNA sequences (VNTRs: 0.1–20 kb).
- **Methodology:** DNA Isolation → Restriction Endonuclease Digestion → Agarose Gel Electrophoresis → Southern Blotting → Radiolabeled VNTR Probe Hybridization → Autoradiography.
- **Applications:** Paternity disputes, forensic crime analysis, and genetic biodiversity studies.

CATEGORY 2: HIGH-FREQUENCY 3-MARK CONCEPTUAL TOPICS

Mechanisms, Differences & Core Proofs

5. Eukaryotic Nucleosome Structure & Chromatin Packaging

HIGH PRIORITY • 3M

- **Histone Octamer:** 2 molecules each of H2A, H2B, H3, H4 rich in basic amino acids (**Lysine & Arginine**).
- **Core Details:** ~200 bp negatively charged DNA wrapped ~1.65 turns around octamer; sealed by H1 histone.
- **Euchromatin vs. Heterochromatin:** Loose/light/active vs. dense/dark/inactive.

6. Semi-Conservative Replication Proof (Meselson & Stahl)

HIGH PRIORITY • 3M

- **Experimental System:** *E. coli* cultured in $^{15}NH_4Cl$ transferred to ^{14}N; evaluated by CsCl gradient centrifugation.
- **Generational Yields:** Gen I (20 min) = 100% hybrid (^{15}N-^{14}N); Gen II (40 min) = 50% light (^{14}N-^{14}N) + 50% hybrid (^{15}N-^{14}N). (Taylor validated in *Vicia faba* with ^{3}H-thymidine).

7. Post-Transcriptional Modifications in Eukaryotes

HIGH PRIORITY • 3M

- **Splicing:** Excision of non-coding introns and ligation of coding exons by spliceosomes.
- **5' Capping:** Addition of 7-methylguanosine triphosphate (m^7Gppp) at the 5'-terminus.
- **3' Polyadenylation:** Addition of 200–300 adenylate residues (Poly-A tail) at the 3'-terminus.

8. Salient Features of the Genetic Code

HIGH PRIORITY • 3M

- **Core Attributes:** Triplet nature (61 sense codons + 3 nonsense codons), unambiguous & specific, degenerate, commaless reading frame, universal nature, and dual role of AUG (initiator + Methionine).

CATEGORY 3: RAPID 2-MARK & 1-MARK HIGH-YIELD TARGETS

Definitions, Chemical Logic & Numbers

9. Rapid-Fire Board Exam High-Yield Points

CORE 2M & 1M

- **DNA vs RNA Stability:** Absence of 2'-OH in deoxyribose + presence of Thymine (5-methyluracil) makes DNA chemically less reactive and structurally more stable than RNA.
- **Chargaff's Equivalence:** [A] + [G] = [T] + [C]; [A]/[T] = 1; [G]/[C] = 1 for double-stranded DNA.
- **Untranslated Regions (UTRs):** Non-coding mRNA sequences flanking the start and stop codons, required for translation efficiency and mRNA stability.
- **Transcription Units:** Flanked by Promoter (upstream 5' relative to coding strand), Structural gene, and Terminator (downstream 3'). Template strand is 3'→5'.
- **Must-Know 1-Markers:** Bacterial ribozyme = **23S rRNA** | Stop codons = **UAA, UAG, UGA** | Largest human gene = **Dystrophin (2.4 Mb)** | Human haploid content = **3.3×10^9 bp** | Bacterial initiation/termination = **σ and ρ factors**.